

Underwater machines

THE FIRST SUBMARINES were simple designs. They allowed travel under water and were useful in war. More modern submarines were powered by diesel or gasoline while on the surface and used batteries under water. In 1955, the first submarine run on nuclear fuel traversed the oceans. Nuclear power allowed submarines to travel great distances before needing to refuel. Today, submarines have sophisticated sonar systems for navigating under water and pinpointing other vessels. They can carry high-powered torpedoes to fire at enemy craft or nuclear missiles. Submersibles (miniature submarines), used to explore the deep seafloor, cannot travel long distances. They need to be lowered from a support vessel on the surface.

Snort mast to renew and expel air with help of bellows

Augur for drilling into enemy ship to attach mine on rope

Vertical propeller

Delayed action mine

Side propeller powered by foot pedals

External steering bar operated by diver

Internal steering position

Hand pump for pressurizing air reservoir and emptying ballast tanks

"TURTLE" HERO

A one-man wooden submarine, the *Turtle*, was used during the Revolutionary War in 1776 to try to attach a delayed-action mine to an English ship blockading New York Harbor. The operator became disorientated by carbon dioxide building up inside the *Turtle*. He jettisoned the mine, which exploded harmlessly. Nevertheless, the explosion was enough to cause the British ship to up anchor and sail away.



UNDERWATER ADVENTURE

Inspired by the invention of modern submarines, this 1900 engraving depicted a scene in the year 2000 with people enjoying a journey in a submarine liner. In a way, the prediction has come true as tourists can now take trips in small submarines to view marine life in places such as the Red Sea. However, most people explore the underwater world by learning to scuba dive or snorkel.

Front wheels smaller than back ones for easier turning